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Strategic mobilization of voters

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ABSTRACT

Organized public demonstrations of voter support for policy issues through rallies and petitions are mechanisms by which interest groups sometimes seek to influence political decision-making. We develop a voter-participation model of an interest group's strategic decision to publicly mobilize supportive voters through either a petition with zero participation cost or through a public rally with positive participation cost. Our model shows that voter mobilization can be influential when elected politicians are sufficiently uncertain about two dimensions of voters' preferences, the breadth of support for the issue and the saliency of the issue. The distribution of voter preferences – defined by the numbers of policy supporters and opposers and election vote-switchers and non-switchers – determines whether low or high participation-cost forms of mobilization are optimal. The model's predictions are consistent with recent mobilization campaigns organized by a range of interest groups, such as firms, environmental activists, and racial justice advocates.

Public protests, rallies, petitions and letter-writing campaigns are staples of liberal democracies because they offer ways for organized interest groups to directly communicate voter preferences to government on specific issues and to potentially influence policy outcomes.[1] Research studies have found that protests, which are costly for voters to participate in, can be effective in shaping elected politicians' policy decisions because they can send a credible signal about voter preferences (Lohmann, 1994; Kollman, 1998). Protests, however, are just one type of communication mechanism, and anecdotal observation indicates that some interest groups rely more heavily on other mechanisms such as petitions, which incur no or minimal participation costs.[2] Uber, for example, has systematically organized online consumer petitions when entering new cities while AirBnb has primarily utilized accommodation owner rallies and demonstrations (Holburn and Raiha, 2017). Hence, how do interest groups determine whether to publicly mobilize voters – and, if so, the optimal type of mobilization – when seeking political influence?

In this paper, we contribute to political economy research by developing a voter-participation model (Battaglini, 2017; Hill, 2022; Lohmann, 1993, 1994) that analyzes the informational conditions when an interest group mobilizes voters either in a zero participation-cost event such as an online petition or else in a positive participation-cost event such as a rally in order to publicly demonstrate to a politician the level of voter support for a policy. Voter participation in a petition or rally provides new information to the politician – who is uncertain about the extent and intensity of voter preferences for a proposed policy – about the expected election impact of implementing the policy, which affects all voters. Voters individually decide whether to voluntarily participate in a

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[1] Surveys of government relations experts indicate that voter mobilization campaigns can be more effective at affecting politicians' policy decisions than lobbying and political campaign donations (Lord, 2000), though the latter tend to be used more frequently by interest groups.

[2] As other contrasting examples, #BlackLivesMatter has heavily used social media hashtag campaigns while Fridays for Future has organized student sit-ins and demonstrations.

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mobilization event: they may or may not support the policy proposal and, importantly, they differ in the intensity of their preferences. For voters with strong preferences, the issue is sufficiently salient to affect their vote for the politician in a future election, while for other voters with moderate preferences the issue is not salient enough to be pivotal in their election vote.

Our model predicts that an interest group will mobilize voters to demonstrate support for a policy when several conditions are satisfied. First, the politician needs to be sufficiently uncertain about the distribution of voter preferences for the policy, specifically about the numbers of vote-switchers and of supporters and opposers. Greater political skepticism about voter support for the policy leads to a higher threshold for participation in an event such as a rally or petition in order to convince a politician to implement the policy. If, on the other hand, the politician is perfectly informed about voter preferences then there is no informational benefit from organized mobilization of supporters.

Second, the preference distribution of voters – the number of supporters and the intensity of preferences – is crucial in shaping the effectiveness of different types of organized events. A zero participation-cost (cheap talk) petition can be influential if there is a large number of voters in a jurisdiction with moderate intensity preferences for the policy issue: a petition that garners a large number of signatures can persuade the politician both that there is broad support for the policy and also that there is likely little opposition. A high participation-cost (costly signaling) rally can be effective when there is a smaller set of voters with sufficiently intense preferences (meaning they will switch their vote for the politician depending on whether the politician implements the policy). If the number of supporters accounts for too small a fraction of voters then neither petitions nor rallies will be feasible, irrespective of the intensity of preferences. However, the interest group does not need a majority of voters to attend a rally in order to positively impact the politician's decision. It is possible that even if a majority of voters opposes the policy – and the politician is aware of this fact – the politician will still implement the policy since it is the intensity of voter preferences that matters to the politician. Because attending a rally is a costly exercise for voters, participation sends a credible signal to politicians about the number of voters who are prepared to switch votes, and a small number may be sufficient to convince the politician.

Our model contributes to political economy research on interest group political influence strategy and the optimal design of activism. We provide new insights into the strategic calculus behind different types of voter mobilization campaigns, an issue that to date has received little scholarly theoretical attention. Even though a few papers examine the rationale for voter participation in public events, the existence of such events in these models is either assumed to be exogenous (e.g. Lohmann, 1993) or is assumed to be a costly-participation protest (e.g. Kollman, 1998). Our model endogenizes both the organization and type of public event, and delineates the key factors that influence an interest group's strategic mobilization decisions. Integral to the interest group's calculus is the model's inclusion of two dimensions of voter preferences – preference for the focal issue and preference for other policy issues. While issue preference and salience have been examined in other studies (Aragones et al., 2015; Hill, 2022), our model is the first to formalize how heterogeneity in the intensity of voter preferences impacts the optimal design of mobilization events and when politicians decide to implement policies that affect a minority intensely or a majority moderately.

The remainder of the paper is organized as follows. In the next section we review political economy literature on voter participation, information aggregation, as well as strategic constituency-building. In the third section we present a model of voter mobilization that includes voters, a political decision-maker, and an interest group that may organize an event. The fourth section presents the model's main results – the focal equilibrium profile of the game – as well as a description of each actor's strategic behavior. The fifth section discusses predictions arising from the model and how they inform our primary research questions. We consider extensions of the model and future research directions before concluding.

1. Literature

Our model relates to several theoretical literatures in political economy. First, we build on models of how voter participation in mass public demonstration events can influence policy-makers' decisions and policy outcomes through an information aggregation mechanism. In one of the earliest papers to address this issue, Lohmann (1993) models the decision of a politician who seeks to maximize voter welfare by matching a policy choice with the state of the world, about which both the politician and voters are uncertain. The model establishes conditions under which voluntary voter participation in a public protest reveals information about the state of the world that influences the politician's decision. In a related model without a political decision-maker (Lohmann, 1994), voters use participation in a protest event prior to a referendum as a communication device that enables them to make inferences about the state of the world. Participation can lead to better informed collective referendum choices by voters, but voters with extreme preferences can also create 'noise', depleting the informational value of mass participation. Later work by Hill (2022) extends these ideas to examine the role that preference intensity plays in shaping election platforms and voter participation in costly pre-election actions. Battaglini (2017) and Banerjee and Somanathan (2001) also extend this research by characterizing the theoretical conditions under which costless protest and demonstration events act as information aggregation mechanisms, specifically the conditions around the precision of individual signals received by voters about the state of the world, the degree of preference conflict with the policy-maker, and the size of the population.

While these models examine the informational role that participation in protest events plays in the strategic behavior of voters and policy-makers, the events themselves are assumed to exist exogenously. By contrast, in our model, we explicitly incorporate a strategic interest group that decides whether to organize an event to mobilize voters in an attempt to sway a policy-maker, and, if so, what type of event to organize.

A second stream of related research examines the optimal design of activism by voters, interest groups, or other economic agents. Most related to our model is recent work by Battaglini and Palfrey (2024), which analyzes a general class of costly collective action games, where a (potentially large) set of agents individually decide whether to contribute to a collective goal, and where there

is imperfect information about individual costs of contribution. Battaglini and Palfrey (2024) show how organizations that cannot make transfers or coerce agents and that rely on voluntary participation – termed *weak mechanisms* – can achieve successful collective action even in very large groups. They also provide conditions under which such volunteer-based organizations emerge. Our model also examines interest groups that can only employ weak mechanisms, though we differ by incorporating collective action that is costless for participants and we examine conditions that shape the interest group's design choice between costly and costless voter participation events.

Morris and Shadmehr (2023) examine the optimal design of activism in a similar setting where citizens are potentially engaged in regime change (i.e. revolution) and where involvement is costly, but where the leader/organization endogenously makes transfers to citizens to compensate for their efforts. Other related papers model interest group activism targeted against a firm or industry (as opposed to a policy-maker) and strategic choices about the type of activism campaign (Abito et al., 2016; Heyes et al., 2018). However, these papers do not model voters as strategic agents, which we do. Aside from these distinct model features, we provide a connection between the literature on strategic activism and the literature on special interest group politics.

A third stream of research in political economy directly examines how mobilization of voters – 'outside lobbying' – can be used as a political influence tactic by interest groups (Kollman, 1998; Baron, 1999, 2001, 2018; De Borger and Glazer, 2015; Wolton, 2021; Gause, 2022). Kollman (1998) develops a signaling model of outside lobbying wherein an interest group can launch a costly voter campaign that signals the interest group's private information. In Baron's models, voter mobilization is also a costly action for the interest group – specifically a firm – that lowers the cost of obtaining legislator votes on legislation affecting the firm. De Borger and Glazer (2015) adopt a similar premise in their model of worker employment, compensation and political mobilization, proposing that by recruiting more workers and offering above-market wages, firms can induce workers to voluntarily mobilize, which in turn reduces the risk of unfavorable regulation being implemented. Wolton (2021) models how interest groups can wield both inside and outside lobbying, where the latter is modeled as having a direct effect on the probability of legislation passing. Finally, Gause (2022) extends the model of Kollman (1998) to examine how resource constraints faced by interest groups affect the efficacy of outside lobbying as credible costly signals, finding that lower resource groups are advantaged in outside lobbying.

In all these models, the strategic decision-making processes of voters and citizens regarding participation are left unmodeled. The primary decision of an interest group is to optimize over an amount of voter mobilization or outside lobbying, which is assumed to deterministically translate into political support or policy outcomes. Voters are assumed to act non-strategically, simply responding to the interest group's directions. By contrast, our model incorporates strategic behavior of both voters and the politician under conditions of uncertainty about the distribution of voter preferences. Further, we endogenize the type of mobilization activity, which is not considered in extant literature.

Fourth, at a more general level, our model contributes to political economy models of interest group influence on policy-makers in contexts where interest groups employ strategic intermediaries or outside forces to influence policy outcomes. For instance, Lyon and Maxwell (2004) show how firms can covertly subsidize other groups to 'astroturf' lobby on their behalf. Raiha (2018) shows how firms can influence voter inferences through changing economic conditions in a jurisdiction through costly capital investments. And papers by Hall and Deardorff (2006) and Schnakenberg (2017) show how interest groups can lobby allied legislators, who in turn lobby other members of a legislature to develop a majority coalition.

2. Model

The model involves three sets of players: an interest group (which we label as a firm), an incumbent politician, and a discrete set V of $N \in \mathbb{N}$ voters.[3]

There is a policy issue determined by the politician, who chooses between two alternatives, $d \in \{0, 1\}$, where $d = 1$ indicates the outcome that is preferred by the firm. The politician does not know (ex ante) the preferences of voters on the policy issue, and thus is uncertain about how the decision will affect her subsequent electoral prospects. However, the politician can potentially make inferences about voter preferences by observing the mobilization actions of the firm and voters, both of whom possess more information about preferences than the politician.

The firm has an opportunity to mobilize its supporting voters (such as its employees or customers), in an effort to convince the politician that she should implement the firm's preferred policy. The firm can organize a single public event that its supporters may voluntarily participate in – $e \in \{0, 1\}$ where $e = 1$ denotes that an event is organized. An event is characterized by the cost of voter participation, $c \in \mathbb{R}_+$. For instance, it is more costly for a voter to attend a public rally than to sign an online petition. Voter participation in an event can convey information to the politician about the direction of voters' preferences on the issue as well as the intensity of preferences, as indicated by the cost of voter participation in the event. For simplicity, we assume there are two possible event types the firm can organize: an event that is costless for voters to participate in, $c = 0$ (e.g. online petition), or an event that has a positive voter-participation cost, $c = \eta > 0$ (e.g. rally).[4] The costless event is effectively a form of cheap-talk on the part of participants, while the costly event is a form of costly signaling. In each case, the cost to the firm of organizing an event, irrespective of its type, is $k > 0$.[5] The event and the politician's policy decision both occur prior to an election, at which point voters choose whether or not to re-elect the incumbent.

[3] As we elaborate in the Appendix, we assume N is sufficiently but not arbitrarily large.

[4] We use the term *petition* to refer to a costless-participation event and the term *rally* to refer to a costly-participation event throughout the paper.

[5] Although we assume the cost of organizing an event is the same for both event types, the results are unchanged if the cost varies by event, as long as the costs satisfy the subsequent conditions described below.

Table 1
Voter Payoffs.

		Voter's Election Decision	
		$v_i = 1$	$v_i = 0$
Politician's Policy Choice	$d = 1$	$\theta + \psi_i - c\rho_i$	$-c\rho_i$
	$d = 0$	$\theta - \psi_i - c\rho_i$	$-c\rho_i$

Payoffs

Before discussing the precise sequence of moves and players' information sets, it is helpful to first describe the players' payoffs. The firm benefits if the politician implements its preferred policy (i.e. $d = 1$) and it also pays a cost k if it organizes a voter event, so its payoff is simply $d - ek$. We assume the politician is intrinsically policy neutral but aims to maximize the number of votes she receives in the election.

Voter payoffs are more complex. Each voter $i \in V$ has two dimensions of preference: (i) policy issue preference ($\psi_i \in \mathbb{R}$), and (ii) valence/other issue preference ($\theta \in \mathbb{R}$), which for simplicity we assume is the same for all voters. The voters' payoffs are determined jointly by the policy choice of the politician as well as by their event participation decision – which we denote by $\rho_i \in \{0, 1\}$, where $\rho_i = 1$ denotes that the voter participates in the event and $\rho_i = 0$ denotes that the voter does not participate in the event. The voter's election vote choice is $v_i \in \{0, 1\}$, where $v_i = 1$ denotes a vote for the incumbent and $v_i = 0$ is a vote for the challenger. To ensure temporal consistency between the politician's policy decision and voters' subsequent election decisions, we assume that each voter's payoff depends, in part, on their election choice.

If voters have valence preference $\theta > 0$ we assume they prefer to re-elect the incumbent politician rather than elect the challenger in the absence of the policy issue, which is introduced by the incumbent (but is not adopted by a challenger). We normalize the voters' payoffs under a challenger at zero.[6] If the voter has a policy preference $\psi_i > 0$ then the voter prefers that the incumbent politician select the policy alternative $d = 1$, which increases the voter's payoff by ψ_i. If the incumbent does not select this policy (i.e. $d = 0$), we assume the voter's payoff decreases by ψ_i. Hence, if the voter cares more about the policy issue than the valence (i.e. $|\psi_i| > |\theta|$), then the voter's election vote for the incumbent or challenger will switch depending on the incumbent politician's policy decision, d: for a voter who is strongly in favor of the policy issue, the payoff is greater from voting for the incumbent than from voting for the challenger ($\theta + \psi_i > 0$). For a voter who is strongly opposed to the policy issue, the payoff from voting for the challenger is greater than from voting for the incumbent ($0 > \theta - \psi_i$). Alternatively, if the voter cares more about the valence than the policy issue (i.e. $|\theta| > |\psi_i|$), then the voter bases his or her election decision solely upon θ.[7]

Table 1 summarizes the voter's payoff structure in four scenarios, according to which policy is selected ($d = 0$ or 1) and whether the voter elected the incumbent or challenger ($v_i \in \{0, 1\}$). The table shows the payoffs in each scenario. If the voter does participate in an event, this reduces scenario payoffs by the voter participation cost c.

For simplicity and clarity of exposition, we assume hereafter that $\theta > 0$.[8] We denote the profile of voters' issue preferences as $\psi = (\psi_1, \psi_2, \ldots, \psi_N)$. Without loss of generality, we order voters in increasing magnitude of their realized issue preference, ψ_i – that is, $i < j \Rightarrow \psi_i \leq \psi_j$ for any $i, j \in V$. Fig. 1 shows an illustrative voter issue preference profile.[9]

The points on the curve above the x-axis show the voters who support a policy decision $d = 1$ (i.e. voters for whom $\psi_i > 0$), while the points on the curve below the x-axis show the voters who oppose it (i.e. voters for whom $\psi_i < 0$). In the example drawn in Fig. 1, slightly more than half of voters oppose $d = 1$ and prefer the politician to choose $d = 0$. However, when deciding whether to vote for the incumbent politician in the election, voters consider both dimensions of their preferences, which influences which decision is electorally advantageous for the politician.

Fig. 2 illustrates the four different types of voter. Voters in region A strongly oppose a policy of $d = 1$ (i.e. $\psi_i < -\theta$). Since the intensity of their policy preference is sufficiently negative, these voters will vote for the incumbent politician if and only if she chooses $d = 0$. By contrast voters in region D strongly support the policy $d = 1$ (i.e. $\psi_i > \theta$) and will vote for the incumbent if and only if she chooses $d = 1$. We refer to voters in regions A and D as *switchers* (either supportive or opposing) since they switch their vote depending on the politician's decision d. We denote the number of supportive switchers by N^s and the number of opposing switchers by N^o, which depend on ψ and θ. In between these two categories are B and C type voters, for whom $|\psi_i| < \theta$. B and C voters do not have a sufficiently strong preference on the policy issue for it to affect their voting decision. As such, we refer to these voters as supportive or opposing *non-switchers*. Since the incumbent politician wishes to maximize the number of votes she receives she would optimally select $d = 1$ given the profile depicted in Fig. 2, even though a majority of voters prefer $d = 0$, because the number of supportive switchers is greater than the number of opposing switchers (i.e. $N^s > N^o$).

Henceforth, we compactly summarize voter payoffs (from Table 1) through the following formula:

$$(\theta - \psi_i + 2\psi_i d)v_i - ce\rho_i.$$

[6] This normalization is made primarily for mathematical simplification, but it could also reflect voter uncertainty about a potential challenger's ideological position on the issue.

[7] This formulation of payoffs is akin to existing literature that models retrospective or expressive voting. Models commonly include voters who exogenously vote retrospectively or expressively (e.g. Austen-Smith and Banks, 1989; Glazer, 1992) or who rationally choose to vote retrospectively (e.g. Alesina et al., 1993; Lohmann, 1998). One can interpret retrospective voting in our model as the optimal voter strategy in circumstances where voters learn about the preferences of politicians and choose not to vote for politicians who have revealed that they do not share the same policy preference as the voter on an issue.

[8] In the Appendix we show that this assumption is without loss of generality since even if $\theta < 0$, our results are qualitatively unchanged.

[9] For visual ease, we depict the voters' preference profile as a continuous curve rather than a discrete set of points.

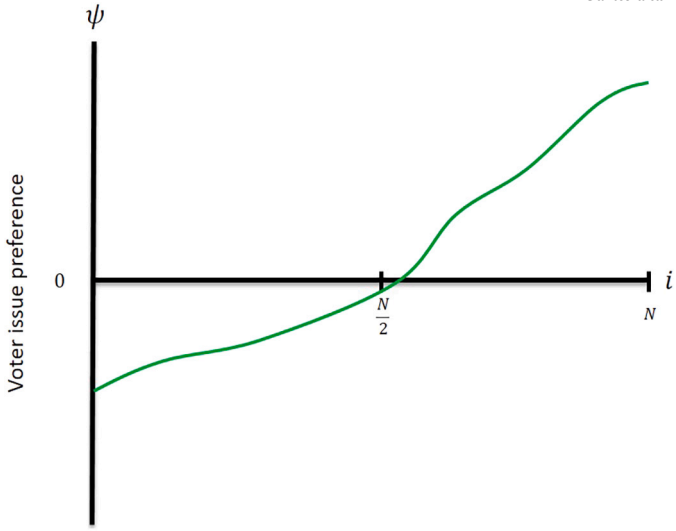

Fig. 1. Illustrative voter issue preference profile.

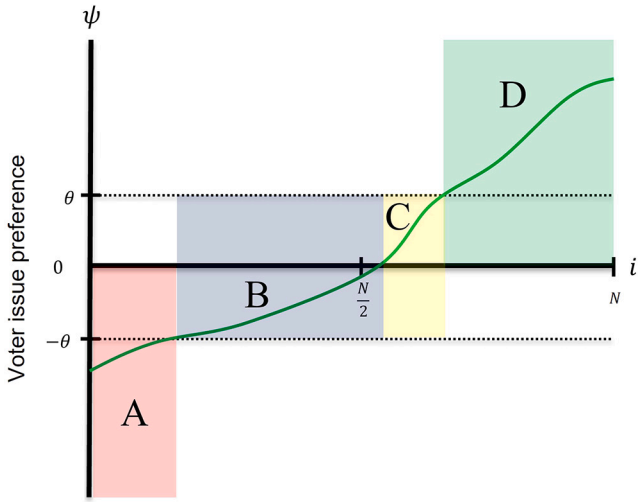

Fig. 2. Categories of voters. A = Opposing switchers. B = Opposing non-switchers. C = Supportive non-switchers. D = Supportive switchers.

Information

If the politician knew precisely the scale and intensity of voter preferences, and hence N^s and N^o, her policy decision would be straightforward, and voter events would not convey new information. However, we assume instead that the politician does not know voters' issue preferences but has prior beliefs. We assume that the voters' preference profile $\psi = (\psi_1, \psi_2, \ldots, \psi_N)$ is stochastic, where each ψ_i is i.i.d. according to the density function F, with $supp(F) = \mathbb{R}$.[10] The value of each ψ_i is known only to voter i, while the politician holds prior beliefs in accordance with the distribution F. We assume F is non-degenerate and also that $P(\psi_i > \theta) < P(\psi_i \leq -\theta)$, which implies that in the absence of new information the politician would optimally choose $d = 0$ (because $E(N^s) < E(N^o)$).

The firm possesses superior but imperfect knowledge of the voters' policy preference profile ψ. After nature determines ψ, the firm receives a signal $\hat{\psi} = (\hat{\psi}_1, \hat{\psi}_2, \ldots, \hat{\psi}_N)$, which with probability π is equal to the realized preference profile $\psi = (\psi_1, \psi_2, \ldots, \psi_N)$, and with probability $1 - \pi$ is equal to a profile $\psi^0 = (\psi_1^0, \psi_2^0, \ldots, \psi_N^0)$ which is stochastic and also distributed according to F. In this way, with probability π the firm correctly observes the preference profile of the electorate, and with probability $1 - \pi$ the firm observes an erroneous preference profile. At this point, we make no assumptions on π, but we will see later how different conditions on both π and the cost of organizing k play a role in mobilization.

Each voter knows his/her own policy preference ψ_i, but does not know other voters' preferences. The common valence preference θ is known to all players.

[10] With appropriate scaling of θ, we could accommodate a bounded support.

Voter Participation and Coordination

Voters voluntarily decide whether to participate in a firm-organized public event, which can convey information about voter preferences that persuades the politician to choose the policy $d = 1$. For an event that is costless to participate in (i.e. $c = 0$), it is easy for voters who support the issue to indicate their support for the policy through participation. But if an event is costly to participate in (i.e. $c > 0$) then even a supporter of the issue would not want to bear the individual cost of participating without a sufficiently high likelihood of being pivotal to the politician's decision. Otherwise, the temptation to free-ride on other voters' participation would inhibit costly participation.

For notational convenience we let $N_p(0)$ denote both the number and set of voters who are supporters – that is $N_p(0) = \{i \in V : \psi_i > 0\}$. Similarly, we let $N_p(\eta)$ denote both the number and set of voters whose preferences are positive and larger than η – that is, $N_p(\eta) = \{i \in V : \psi_i > \eta\}$. Since the firm observes $\hat{\psi}$, which is not necessarily the actual preference profile ψ, we also analogously define $\hat{N}_p(c)$ as the set of perceived voters whose preferences meet the above thresholds according to the firm's observation $\hat{\psi}$. Q denotes the actual number of observed participants – that is, $Q = \sum_{i \in V} \rho_i$.

If the firm organizes a costly event, its supporters would not participate unless they reasonably expected to be pivotal in the politician's decision – otherwise they would likely free-ride (Olson, 1965; Palfrey and Rosenthal, 1983). For modeling parsimony, we assume the firm overcomes the free-riding problem by utilizing a coordinator – a dummy actor who knows voter preferences and who can send signals indicating whether their participation is likely to be pivotal when an event is costly.[11] When the firm organizes an event with a positive cost of participation $c = \eta > 0$, it further instructs the coordinator to mobilize a target number of supporters, which we denote Q^M. Given this target the coordinator mobilizes supporters by sending to each voter a participation signal $\alpha_i \in \{0, 1\}$, which is observed only by the voter who receives it. The coordinator sends the signal $\alpha_i = 1$ only to Q^M of voters in $N_p(\eta)$ if the set of such voters – whose issue preference is sufficiently strong – is at least as large as Q^M. If there does not exist at least Q^M voters in $N_p(\eta)$, then the coordinator simply sends the participation signal $\alpha_i = 1$ to all voters in $N_p(\eta)$. More precisely, given a target Q^M, the coordinator selects a vector $\alpha = \{\alpha_1, \dots, \alpha_N\}$, such that $\sum_{i \in N_p(\eta)} \alpha_i = Q^M$ or $\alpha_i = 1 \ \forall i \in N_p(\eta)$.[12]

Sequence

The sequence of the game is as follows:

1. Nature selects the profiles ψ and ψ^0 according to the distribution F, and sends a signal $\hat{\psi}$ to the firm, for which $\pi = P(\hat{\psi} = \psi)$ and $1 - \pi = P(\hat{\psi} = \psi^0)$.
2. The firm decides whether to organize an event for voters to participate in, $e \in \{0, 1\}$, where $e = 1$ indicates an event is organized, while $e = 0$ indicates no event occurs. The cost to the firm of organizing an event is $k > 0$. If $e = 1$, then the firm selects a cost of participation $c \in \{0, \eta\}$.
3. If the firm organizes an event with positive voter participation cost (i.e. $e = 1$ and $c = \eta$), then the firm sets the mobilization target Q^M for the coordinator. The coordinator sends private signals to each voter, $\alpha = (\alpha_1, \dots, \alpha_N)$, where $\alpha_i \in \{0, 1\}$. If $Q^M \leq N_p(c)$, then the coordinator sends signal $\alpha_i = 1$ to Q^M voters from $N_p(c)$, and otherwise sends the signal $\alpha_i = 1$ to all voters in $N_p(c)$.
4. If an event is organized, each voter decides whether to participate, $\rho_i \in \{0, 1\}$ (where $\rho_i = 1$ indicates that voter i participates).
5. After observing event organization, voter participation, and the participation cost, the politician decides which policy alternative to implement, $d \in \{0, 1\}$.
6. Voters decide whether to re-elect the incumbent politician or to elect the challenger; each voter i chooses $v_i \in \{0, 1\}$ where $v_i = 1$ indicates a vote for the incumbent and $v_i = 0$ indicates a vote for the challenger.
7. Payoffs are realized.

Strategy Space and Solution Concept

We focus on pure strategies, and describe the strategies for each of the players as follows[13]:

- The firm's strategy has three elements: an event decision $e \in \{0, 1\}$, a voter participation cost decision $c \in \{0, \eta\}$, and a mobilization target decision $Q^M \in \mathbb{N}$. These are all determined after observing the signal $\hat{\psi} \in \mathbb{R}^N$. Thus, the firm's strategy is a triple $(e, c, Q^M) : \mathbb{R}^N \to \{0, 1\} \times \{0, \eta\} \times \mathbb{N}$.
- Each voter's strategy has two elements: a participation decision $\rho_i \in \{0, 1\}$ if the firm has organized an event; and a voting decision $v_i \in \{0, 1\}$. In making their participation decision, the voter observes the cost of participation $c \in \{0, \eta\}$, the coordinator message $\alpha_i \in \{0, 1\}$, and their own issue preference $\psi_i \in \mathbb{R}$. In making their voting decision, the voter observes the firm event decision $e \in \{0, 1\}$, the cost of participation $c \in \{0, \eta\}$, the coordinator's message $\alpha_i \in \{0, 1\}$, their participation decision $\rho_i \in \{0, 1\}$, the politician's issue decision $d \in \{0, 1\}$, and their own issue preference $\psi_i \in \mathbb{R}$. Thus, for each voter i, their participation strategy is a function $\rho_i(c, \alpha_i, \psi_i) \in \{0, 1\}$, and their voting strategy is a function $v_i(e, c, \alpha_i, \rho_i, d, \psi_i) \in \{0, 1\}$. Let $\rho = (\rho_1, \dots, \rho_N)$ and $v = (v_1, \dots, v_N)$ denote the profiles of voter participation and voting decisions.

[11] The coordinator performs the same role as the omniscient mediator in Bayes Correlated Equilibrium by providing coordination signals to players (Bergemann and Morris, 2016; Forges, 2023). Pogorelskiy (2014) provides a political economy application of Bayes Correlated Equilibrium in the context of costly voter turnout with pre-play communication.

[12] We show below that the specific value for Q^M will be determined in equilibrium and will be the pivotal cutoff expected by the firm.

[13] For simplicity we use the same symbols to represent actions and strategies.

- The politician's strategy involves the issue decision $d \in \{0,1\}$. In making her decision, she observes the firm's event decision $e \in \{0,1\}$, the event cost of participation $c \in \{0,\eta\}$, and the number of event participants $Q \in \mathbb{N}$. Thus, her strategy is a function $d(e,c,Q) \in \{0,1\}$.

The solution concept we use is perfect Bayesian Nash equilibrium (PBNE), which specifies a profile $(e^*, c^*, Q^{M*}, \rho^*, d^*, v^*, \mu_F, \mu, \mu_P)$ of strategies and beliefs according to the following:

- For the firm, using d^* to anticipate the politician's decision: a decision on whether to organize an event $e^* \in \{0,1\}$ and, if so, the cost of participation $c^* \in \{0,\eta\}$; if $c^* = \eta$, a mobilization target $Q^{M*} \in \mathbb{N}$ for each signal $\hat{\psi} \in \mathbb{R}^N$; and beliefs μ_F about ψ, such that:

$$(e^*, c^*, Q^{M*}) = \arg\max_{e,c,Q^M} E\left[d^* - e \cdot k \mid \hat{\psi}, \mu_F\right].$$

- For each voter i, using d^* and v^* to anticipate the politician's issue and voters' election decisions respectively, and provided $e^* = 1$: a participation decision $\rho_i^* \in \{0,1\}$ for each participation cost $c \in \{0,\eta\}$, coordinator's message $\alpha_i \in \{0,1\}$, issue preference $\psi_i \in \mathbb{R}$, and beliefs μ_i about ψ, such that:

$$\rho_i^* = \arg\max_{\rho_i} E\left[(\theta - \psi_i + 2\psi_i d^*)v_i^* - ce\rho_i \mid e, c, \alpha_i, \psi_i, \mu_i\right].$$

- For the politician, using v^* to anticipate the voters' election decisions: an issue decision $d^* \in \{0,1\}$ for each firm event decision $e \in \{0,1\}$, participation cost $c \in \{0,\eta\}$, number of participants $Q \in \mathbb{N}$ attending an event, and beliefs μ_P about ψ, such that:

$$d^* = \arg\max_{d} E\left[\sum_{i \in V} v_i^* \mid e, c, Q, \mu_P\right].$$

- For each voter i: an election vote decision $v_i^* \in \{0,1\}$ for each firm event decision $e \in \{0,1\}$, cost of participation $c \in \{0,\eta\}$, coordinator's message $\alpha_i \in \{0,1\}$, participation decision $\rho_i \in \{0,1\}$, politician's issue decision $d \in \{0,1\}$, their own issue preference $\psi_i \in \mathbb{R}$, and beliefs μ_i about ψ, such that:

$$v_i^* = \arg\max_{v_i} E\left[(\theta - \psi_i + 2\psi_i d)v_i - ce\rho_i \mid e, c, \alpha_i, d, \psi_i, \mu_i\right].$$

Beliefs μ_F, μ_P, and μ_i for each $i \in V$ follow Bayes' rule on the equilibrium path.

The two different voter payoff functions arise because the type of actions the voters take at each stage and the information they condition on are different.

3. Results

Signaling and cheap-talk models often yield multiple equilibria, requiring evaluation of which ones are most plausible. In our model, there are PBNE where no stakeholder events occur or the politician does not implement the firm's preferred policy, as well as equilibria where stakeholder events occur only some of the time when they would be beneficial for the politician. Many such equilibria rely on unrealistic off-equilibrium path beliefs to be sustained.[14] We focus our attention on the PBNE that satisfies an extended version of the intuitive criterion refinement (Cho and Kreps, 1987), where joint deviations by the firm and its supporters are possible. In the Appendix we further detail that the PBNE we focus on is the unique PBNE that satisfies this refinement within a broad plausible class of strategy profiles – namely profiles where the politician employs threshold strategies.[15] As such, we focus only on strategy profiles where the politician employes threshold strategies. This PBNE is also the profile that is the best for the firm in expectation. We now describe the focal equilibrium of the game and examine the conditions under which the firm mobilizes voters in order to influence the politician's policy decision. The formal description of the equilibrium profile is as follows:

Proposition 1. *If $\pi > k > 1 - \pi$, then there exist thresholds $Q^*(c)$, such that the following strategy profile is a PBNE of the game:*

Firm: (i) *If $\hat{N}_p(0) \geq Q^*(0)$, then select $e^* = 1$ and $c^* = 0$.*
(ii) *If $\hat{N}_p(\eta) \geq Q^*(\eta)$ and $\hat{N}_p(0) < Q^*(0)$, then select $e^* = 1$, $c^* = \eta$, and $Q^{M*} = Q^*(\eta)$.*
(iii) *Otherwise select $e^* = 0$.*

Voter Participation: *Choose $\rho_i^* = 1$ if either (i) or (ii) below hold. Otherwise, choose $\rho_i^* = 0$.*
(i) *If $e = 1$, $c = 0$, and $\psi_i > 0$.*
(ii) *If $e = 1$, $c = \eta$, and $\alpha_i = 1$.*

[14] For example, if the politician were to always attribute event participation as being done by opposing switchers, this could support an equilibrium wherein mobilization does not occur even though there are circumstances when it would benefit the voters, firm, and politician.

[15] I.e. the politician chooses $d = 1$ if and only if $Q(c) > \bar{Q}(c)$ for some $\bar{Q}(c)$ thresholds. Threshold strategies are commonly the focus of attention in this literature (e.g. Lohmann (1993), Battaglini (2017)).

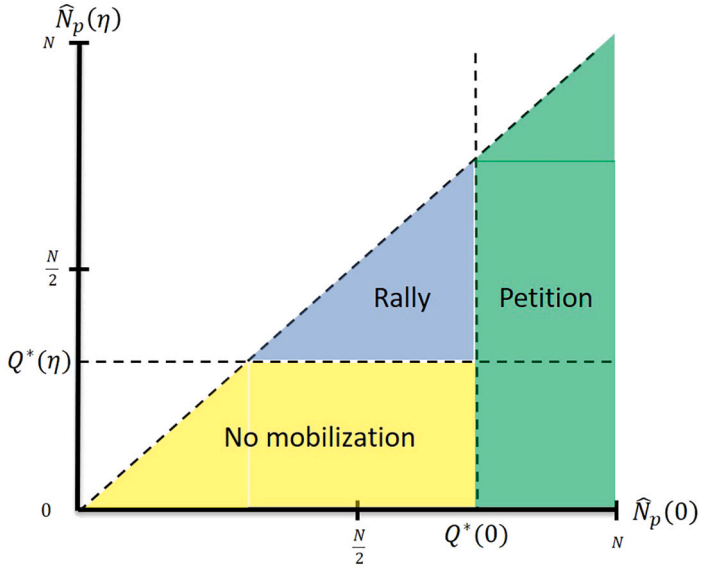

Fig. 3. Mobilization outcome regions. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

Politician: *If $Q \geq Q^*(c)$, then choose $d^* = 1$. Otherwise choose $d^* = 0$.*
Voter Election Decision: (i) *If $|\theta| > |\psi_i|$, then vote $v_i^* = 1$.*
(ii) *If $|\theta| \leq |\psi_i|$ and $\psi_i \geq 0$, then vote $v_i^* = 1$ if $d = 1$, and vote $v_i^* = 0$ if $d = 0$.*
(iii) *If $|\theta| \leq |\psi_i|$ and $\psi_i < 0$, then vote $v_i^* = 1$ if $d = 0$, and vote $v_i^* = 0$ if $d = 1$.*

Beliefs follow Bayes' rule where possible. In the single off-equilibrium event that $c = \eta$ and $Q > Q^*(\eta)$, the politician forms the same beliefs about voters as in the on-equilibrium $Q = Q^*(\eta)$ case.

Intuitively, the thresholds $Q^*(c)$ represent the minimum number of participants the politician would need to observe in order to believe that there is a greater number of supportive switchers than opposing switchers (see Appendix for details).

There are two key conditions that are essential for the above PBNE. First, in assuming $\pi > k > 1 - \pi$, we assume that the firm's information is more likely correct than incorrect – that is, $\pi > 1 - \pi$. Placing the cost of organizing k in between ensures that the firm will only organize an event if it expects to succeed in persuading the politician.[16]

A second, less intuitive, condition involves what occurs by chance if the firm's information is incorrect (see Appendix for details).

Firm. The firm's equilibrium strategy depends upon expected stakeholder participation in an event, as determined by the signal it receives about the voters' issue preference profile, compared to the politician's voter participation thresholds, $Q^*(c)$. The firm will organize an event only if its private signal about voter preferences indicates there is a sufficiently high number of voters willing to participate in an event that meets the politician's threshold for implementing the policy. If the firm expects there will be insufficient support, it will not pay the cost of organizing an event.[17] The firm's decision on the type of stakeholder event to organize is determined by the expected number of willing participants for each type, $\hat{N}_p(0)$ and $\hat{N}_p(\eta)$.

Fig. 3 illustrates how mobilization outcomes vary with different realizations of $\hat{N}_p(0)$ and $\hat{N}_p(\eta)$. If there is insufficient weak support for a petition but sufficient strong support (i.e. among vote-switchers), the firm will organize a rally to demonstrate narrower but deep support for the policy proposal (i.e. if $\hat{N}_p(\eta) > Q^*(\eta)$ and $\hat{N}_p(0) < Q^*(0)$) – the blue region in Fig. 3. If the rally is sufficiently costly for voters to participate in, the politician will infer that each participant is a supportive switcher for whom the issue will decide their election vote. Moreover, since the firm organizes participation in the rally by precisely the threshold number of voters – so that each is pivotal – the politician will also infer that there are other supportive switchers beyond the observed participants of the rally whose vote will depend on the politician's decision.

By contrast, when there is a sufficiently large group of supporters (either weak or strong) – specifically, if $\hat{N}_p(0) > Q^*(0)$ – the firm will organize a zero participation cost event, such as an online petition, to convey the breadth of support for the firm (the green region in Fig. 3). Even if the politician believes that only a fraction of the petition signatories are strong supporters of the policy (i.e. supportive switchers), as long as the number of signatures is sufficiently high the politician will revise upwards her beliefs about the

[16] Since the payoffs to the firm from the policy being approved or not are normalized to one and zero, respectively, the assumed bounds on k are simplified. If the payoff to the firm from the policy being approved were instead B then we would assume that $B\pi > k > B(1 - \pi)$.

[17] There is a qualitatively similar equilibrium when the cost of organizing is $k = 0$, wherein the firm at the very least organizes a costless event. A similar intuition applies to that case.

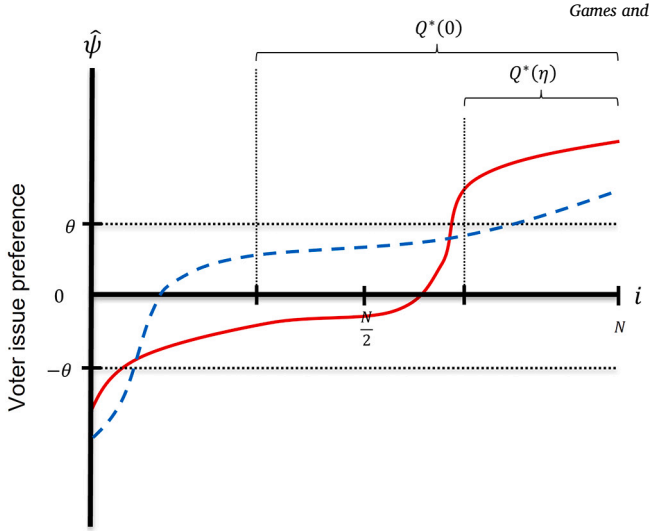

Fig. 4. Alternative voter preference profiles.

number of supportive vote-switchers, and revise downwards her beliefs about the number of opposed vote-switchers, leading her to implement the policy.

Fig. 4 illustrates how distinct and contrasting realizations of $\hat{\psi}$ can result in different mobilization activities. The blue (dashed) curve depicts a distribution profile where a majority of voters prefers the policy proposal, but the support is 'shallow' – the preference intensity is below the vote-switching limit. Out of this majority, there is a small number of voters with strong supportive preferences who are vote-switchers on the policy issue (shown by the segment of the blue (dashed) curve that is above the upper dotted horizontal line). There is a minority of voters with opposing preferences – a small number with weak opposing preferences and a small number with strong opposing preferences. For expositional purposes, we refer to this as an 'Uber' type distribution where weak supporters can be interpreted as actual and potential customers (who moderately prefer the price and convenience attributes of ride-sharing to traditional taxi service), strong supporters as Uber drivers, and opposers as taxi-industry drivers and companies. In this example, the number of strong supportive voters (vote-switchers) is less than the politician's required threshold for a rally, but the total number of supporters exceeds the threshold for a petition. The firm will hence organize a petition, which will be signed by all supporters, causing the politician to infer that the number of supportive vote-switchers is larger than the number of opposing vote-switchers if she approves the issue. Uber has in fact regularly organized petitions of its riders and drivers – which are easily implemented through its app – to illustrate support for ride-sharing, in some cases achieving almost one million signatures (Holburn and Raiha, 2017).

By contrast, the voter distribution profile depicted by the red (solid) curve has only a minority of voters that supports the policy, but most of these supporters have strong preferences for the issue (i.e. the portion of the red (solid) curve above θ). A majority of the voters are in fact weak opposers, with a very small number of opposing switchers. In this case, the number of opposing vote-switchers is smaller than the number of supporting vote-switchers. We refer to this as an 'AirBnb' type of distribution – a small number of travelers (who are likely to reside outside the electoral district) and local home owner-renters are loyal clients of AirBnb while a large number of voters in the district are somewhat negatively affected by impacts on local rental markets, property values and neighborhood environments. In this case, the number of supporters falls far short of the politician's threshold for a petition but the number of strong supporters exceeds the threshold for a rally, which the firm is able to successfully organize. Notably, AirBnb has commonly organized home-owner public rallies at local city halls and legislatures as part of its political advocacy strategy in various jurisdictions – but not petitions (e.g. Said (2015), Noto (2018)).

Although mobilizing voters through costly participation events such as rallies conveys credible information to politicians on the minimum number of vote-switchers on an issue, petitions can also be informative, depending on the distribution profile of voter preferences. In the example in Fig. 5, all voters support the policy issue. Most of the voters are weak supporters, with a small fraction being strong supporters. Implementing a petition – which would be signed by all voters – enables the firm to credibly demonstrate to the politician that there is no opposition whatsoever to the policy. While in this example the number of strong supporters is less than the politician's threshold for a rally, even if it exceeded the threshold, the firm would still prefer to organize a petition rather than a rally: the former would conclusively demonstrate the absence of opposition while the latter would still leave the politician uncertain about the number of strong opposers. Admittedly, this is an extreme scenario. However, it illustrates the informational value to a politician of a petition: as the number of petition signatories increases, the number of possible opposers to the policy issue correspondingly shrinks, causing politicians to lower their expectations of the number of opposing vote-switchers.

We note that the equilibrium profile in Proposition 1 is a PBNE for a certain set of model parameters. However, for some parameter values there also exists a very similar PBNE wherein if the firm believes either type of event is likely to garner sufficiently high

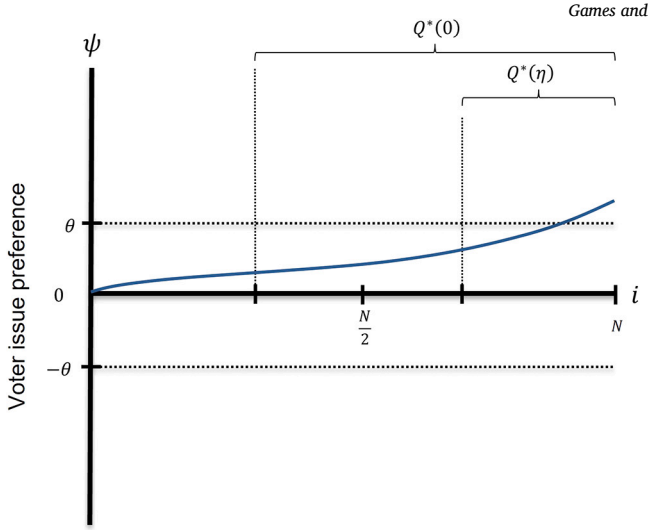

Fig. 5. Extensive, shallow voter support.

participation, then the firm chooses to organize a costly event rather than a costless event. Since this profile is similar to the one above we present the full equilibrium profile in the Appendix. [18]

Politician. The politician observes whether a stakeholder event e occurs and, if so, the cost of voter participation, c, and the level of participation Q. Based on her observations of e, c, and Q, the politician updates her beliefs about the distribution of voter preferences over the issue. In equilibrium, the politician employs a threshold strategy wherein she chooses to implement the policy $d = 1$ if and only if she observes a sufficient level of voter participation in an event. The thresholds for a petition and rally are $Q^*(0)$ and $Q^*(\eta)$, respectively. Each threshold is defined as the lowest number of observed participants, Q, such that the politician believes there is a greater number of supportive switchers than opposing switchers. As noted above, we assume that the politician's prior belief – i.e. the distribution F – is that there is greater opposition to the policy than there is support. However, if a large enough group of voters publicly participate in an event, this can cause the politician to update her beliefs about voters' issue preferences and hence implement the policy.

Since, as demonstrated in Fig. 3, the thresholds $Q^*(0)$ and $Q^*(\eta)$ are the key elements that determine the mobilization activities of the firm, and whether or not mobilization occurs, we now present several propositions that describe comparative statics and that show how the thresholds change with the model parameters.

Proposition 2. *For any η, γ where $0 < \eta < \gamma$ then:*

1. $Q^*(0) > Q^*(\eta)$
2. $Q^*(\eta) > Q^*(\gamma)$

This result shows that equilibrium thresholds are decreasing in the cost of participation, so if an event is more costly for a voter to participate in, fewer participants are needed to persuade the politician to choose $d = 1$. The threshold for a costless event is greater than for a costly event since the set of willing participants, $N_p(0)$, contains more voters than just supportive switchers. For instance, supportive voters would be willing to sign an online petition but only a fraction of them would hold preferences strong enough to switch their vote based on the issue. The politician requires a greater level of participation in a costless event than in a costly event.

The politician's optimal participation thresholds account for the firm's equilibrium choices, namely (i) whether an event is organized, and (ii) the type of event. When the politician observes an event being organized, since the cost of organizing is non-zero, the politician can infer that the firm observed a sufficiently optimistic signal about the voters' issue preference profile. However, the politician also knows that the firm's information could be incorrect, which is why it is essential that she observe the actual voter participation level, in part to verify whether the firm's information was correct or not. If the politician observes the firm organizing an event with a positive voter participation cost, such as a rally, then she knows that the firm observed $\hat{N}_p(0) < Q^*(0)$ – that is, the firm believes there are too few willing participants for a costless event to succeed. This is accounted for in the politician's equilibrium threshold $Q^*(\eta)$.

[18] When there is sufficient support for both types of event, the firm organizes the single event that it expects is most likely to succeed in the situation that its observed profile $\hat{\psi}$ is incorrect. The focus on the scenario of an incorrect profile is to due the fact that the firm is indifferent between the two types of events when its observed profile is correct (since both succeed with certainty), but each event may have a different likelihood of succeeding purely by chance. One advantage of the costless event is that all supporters of the issue, regardless of the intensity of their preference or likely pivotality, are willing participants. This can make it more likely that an event succeeds purely by chance, even if the firm's information is not correct.

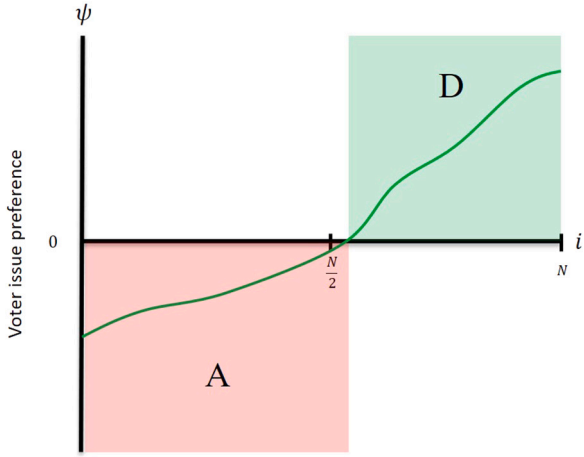

Fig. 6. Categories of voters when $\theta \to 0$.

Proposition 3. *If $\theta \to 0$, then:*

1. $Q^*(0) \to \frac{N}{2}$
2. $Q^*(\eta) \to \frac{N}{2}$
3. *The firm will only organize petitions (i.e. costless events).*

This proposition shows that the valence dimension of voter preferences, θ, is crucial for generating variation in the type of mobilization event selected by the interest group. If θ is removed as a meaningful consideration (i.e. as $\theta \to 0$), mobilization collapses to a petition and there is no consequential variation in the equilibrium choice of the firm. The politician's thresholds converge to $\frac{N}{2}$ because all voters effectively become switchers on the issue, and the politician's goal is to determine whether supporters or opposers constitute the majority of the electorate, as Fig. 6 illustrates (the set of non-switchers vanishes). This finding demonstrates that in standard models in the literature a simple petition would suffice to demonstrate to the politician the level of support, and endogenously would be the only type of mobilization campaign organized by the firm. Variation in the type of mobilization event, therefore, results from the inclusion of the valence dimension, which causes voters to consider the relative importance of the issue, creating a category of voters who are non-switchers. Conveying the intensity or depth of support becomes an important strategic choice for the firm because this information can persuade the politician about the numbers of supportive and opposing switchers.

Proposition 4.

1. *If $P(\psi_i > \theta)$ decreases, then $Q^*(0)$ and $Q^*(\eta)$ increase.*
2. *If $P(\psi_i \leq -\theta)$ increases, then $Q^*(0)$ and $Q^*(\eta)$ increase.*

This proposition shows how changes to the distribution F and the prior beliefs of the politician can shift the politician's thresholds of voter participation. If the politician is naturally more skeptical in that she is either (i) less likely to believe a supporter is a supportive switcher, or (ii) more likely to believe an opposer is an opposing switcher, then the politician requires a greater burden of proof in the form of higher participation.

Fig. 7 depicts the impact of the politician's prior beliefs about voter preferences on the politician's participation thresholds in two scenarios involving different priors. In the first scenario, the politician's beliefs result in petition and rally participation thresholds represented by the dotted vertical black lines. We can see that, given the actual realized distribution of voter preferences – as shown by the green (solid) curve – there are sufficient participants for the firm to organize either a petition or a rally. In the second scenario, the priors are such that either $P(\psi_i > \theta)$ is lower, and/or $P(\psi_i \leq -\theta)$ is higher – that is, the politician is more skeptical that voters are supportive of the policy proposal. These different beliefs lead to the politician requiring higher levels of voter participation in events in order to sufficiently update prior beliefs for her to choose $d = 1$ – these different thresholds are represented by blue dashed vertical lines located to the left of the black dotted lines in Fig. 7. In this new situation, there are not enough supporters to reach the petition threshold but there are sufficient supportive switchers to justify a rally. These two scenarios show how the optimal choice of event type can change due to differences in prior beliefs, even though the underlying realized preference profile $\hat{\psi}$ is the same.

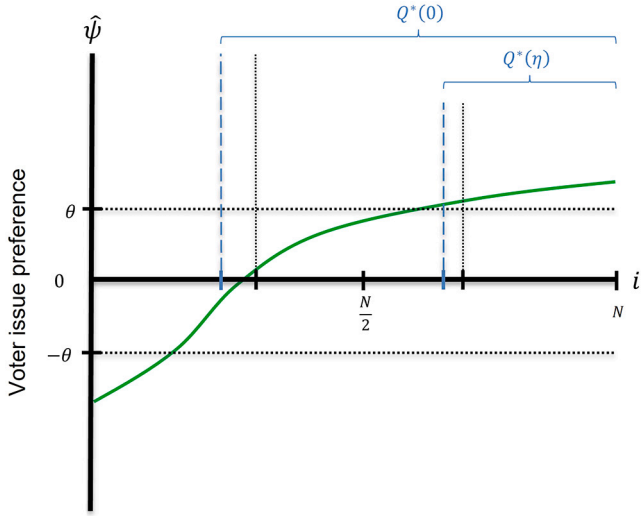

Fig. 7. Politician prior beliefs and optimal mobilization tactics.

Voters. The voters' equilibrium strategies are more straightforward than the strategies of the politician or firm. In the final stage of the game, after the politician has made a policy decision, d, the non-switchers automatically vote for the incumbent politician. Supportive switchers vote for the politician if $d = 1$ and the challenger if $d = 0$, while opposing switchers do the opposite.[19]

If the firm organizes an event, voters must also decide whether to participate. When the event is costless to participate in, voters participate if and only if they support the policy (i.e. $\psi_i > 0$). All voters who support the policy will participate, regardless of the intensity of their preferences. This increases the probability that an event succeeds purely by chance, even if the firm's information about the distribution of voter preferences is not correct. The voter's participation decision for a costly event is more complicated due to potential collective action problems. For a voter to bear the cost of participating, she requires some assurance that participation will be pivotal in convincing the politician to choose $d = 1$. When the firm chooses to organize a costly event, it signals its information about voters' policy preferences (i.e. $\hat{\psi}$) to voters, which tells the voter that as long as the firm's information is correct (which it is with probability π) then the event will succeed. The coordinator organizes voters to ensure that no surplus effort is wasted by inviting only a pivotal set of willing voters to participate.

Because of the underlying uncertainty about the voters' preference profile ψ, a feature of the equilibrium presented in Proposition 1 is that a variety of outcomes are possible along the equilibrium path. For example, it is possible the firm may mistakenly forgo organizing an event even though it would profit from doing so (e.g. when $\hat{\psi}$ is incorrect and more pessimistic than ψ), or it may organize an event that fails to persuade the politician (e.g. when $\hat{\psi}$ is incorrect and more optimistic than ψ). It is possible that voters may participate in events and find their efforts are in vain (i.e. the event fails to garner sufficient participation to persuade the politician). Moreover, the final decision made by the politician may not even be the electorally optimal one: even though there are more supportive switchers than opposing switchers, the number of supporters is insufficient to change the politician's beliefs. On the converse, there are circumstances where even if there are more opposing switchers than supportive switchers, since the firm only demonstrates support (rather than opposition), the stakeholder event may convince the politician to implement the policy ($d = 1$), leading to a reduction in votes. Hence, the equilibrium in our model is consistent with the wide variety of voter event outcomes that are observed empirically.

Information, Issue Salience, and Mobilization. While it is clear from the information structure of our model that uncertainty about voter preferences is essential for voter mobilization to be deployed by interest groups, there are two further predictions about how the firm's information accuracy and issue salience affect the use of mobilization.

Proposition 5. *If $k > \pi$ and $1 - \pi > \pi$, then there is no PBNE where $e = 1$ for any realization of $\hat{\psi}$.*

This proposition shows that if the firm's information is known to be unreliable – both to itself and others – and the cost of organizing an event is sufficiently (though modestly) high, then the firm will not organize an event of any type. Hence, voter mobilization is more likely to be employed by interest groups who have better information about their supporters' preferences than politicians possess. For example, app-based technology companies have detailed information about customers' locations and usage profiles, enabling

[19] If alternatively we assume $\theta < 0$, the optimal voting decision of supportive and opposing switchers is the same as for $\theta > 0$. The optimal voting decision of non-switchers changes such that they vote for the challenger rather than the incumbent. This does not change the optimal strategies of the politician or firm because the optimal voting strategy for non-switchers is unchanged by the policy decision d, and the firm and politician are only concerned with the decisions made by switchers, whose optimal strategies change depending on the decision d. See Appendix, Proof of Proposition 1, for details.

them to gauge the scale and extent of potential support within a politician's electoral district and hence to predict participation in an organized mobilization event.

Proposition 6. *If θ increases, then $P(e = 1)$ decreases – that is, as θ increases, the probability of observing any event decreases.*

This proposition states that if the salience of the issue (as captured by the magnitude of ψ_i relative to θ) decreases, then the firm mobilizes voters less frequently. Firms and other interest groups are more likely to mobilize voters for issues that are particularly important and for which there are potential vote switchers. If there are unlikely to be switchers, then the mobilization effort is less likely to sway the politician.

Many policy issues are not sufficiently salient or important for voters to induce vote-switching based on the politician's policy stance. For example, minor changes in fiscal policy or government procurement standards are unlikely to induce vote-switching behavior among voters, making organized voter mobilization events about such issues less valuable and less likely to be observed.

This result also helps explain the puzzle of why voter mobilization, which is deemed by practitioners to be a particularly effective influence tactic, is seemingly under-utilized by politically active firms, in comparison to lobbying activities or campaign contributions (Lord, 2000). Most regulatory and policy issues that affect firms are not election issues that will sway voters, and most policy debates are likely to concern existing issues around which politicians already understand the fault lines of voter support and opposition. As such, the set of opportunities for rallies and petitions to be valuable for politicians in informing their policy decisions may be limited.

4. Discussion

The previous section presents the focal equilibrium of the game where an interest group decides whether to mobilize voters in a public event in an effort to persuade a politician to implement a policy when the politician is uncertain about voter preferences and the subsequent impact of the policy on future election outcomes. Voter voluntary participation in events can convey credible information about voters' preferences – specifically the breadth and depth of support for a policy – which is otherwise unknown to the politician. When voter events such as public rallies are costly to participate in, they indicate not only the direction but also the intensity of voter preferences on an issue. This information is valuable for politicians (but cannot be readily gathered from a poll) since it is the number of election vote-switchers on an issue – those who sufficiently care about the issue to base their vote upon it – who influence election outcomes. Although the interest group is presented as being a firm in the model, it could instead be an NGO, social advocacy group, or labor union that is considering organizing its supporters in order to influence a policy decision.

Our model provides predictions about the conditions under which interest groups are likely to publicly mobilize voters in order to influence policy outcomes. First, when politicians are uncertain about voter preferences, mobilization events can provide information to the politician about the extent and depth of support for a policy issue. Political uncertainty is greater for novel policy issues, which may explain why stakeholder mobilization has been a common tactic employed by firms such as Uber, Airbnb, Tesla, as well as by medical cannabis companies and in sectors where new regulations are required – and where voter preferences are uncertain (Holburn and Raiha, 2017). Even in established industries or policy arenas, politicians may be uncertain about voter preferences in specific geographic areas or demographic sub-groups, creating an opportunity for targeted stakeholder mobilization efforts to be beneficial. Public mobilization campaigns can also be especially persuasive when they are organized in response to new and emerging social issues.[20] For instance, when police brutality became a salient issue in 2020 after the death of George Floyd in Minnesota, the Black Lives Matter movement initiated a wave of marches across the country to demonstrate the depth of voter support for legislated reforms to police practices.

Even though the illustration of the model that we present is of an interest group (specifically, a firm) marshalling its supporters to influence the policy decision of an elected politician, the model can be applied more generally to other contexts where the policy decision-maker is an entity other than a politician. For example, the model can also be interpreted as describing the mobilization strategy of an environmental NGO considering whether to launch a public campaign against a firm, in order to convince the firm to change specific environmental practices. The targeted firm may be uncertain about the extent and intensity of broader customer support for the NGO's campaign demands, and hence may be susceptible to customer demonstrations organized by the NGO. Customer petitions, letters or public demonstrations can convey credible information to the firm about the scope and intensity of customer preferences for its campaign. While customers do not 'vote' in an electoral sense, they do express their preferences through their buying decisions, potentially switching from the targeted firm to a competitor. In this setting, the policy decision-maker in the model is the firm, which must decide whether to implement the NGO's policy demand after observing customer participation in an NGO-organized campaign and making inferences about the distribution profile of broader customer preferences. This flexibility in interpreting our model expands its relevance for understanding a variety of phenomena involving collective voluntary participation, including those outside conventional political settings.

[20] For example, the environmental NGO Fridays for Future, started in 2018 by activist Greta Thunberg, organized high-school students to strike on Fridays (i.e. not attend school), and instead participate in demonstrations at local government legislatures in support of environmental policy changes to combat climate change. The environmental policy preferences of youths (i.e. future voters) are conveyed through such campaigns.

5. Conclusion

In this paper, we develop a model that examines the conditions under which an interest group strategically mobilizes voters in a public event in order to convey information to a politician about the breadth and depth of voter preferences on a policy issue, thereby influencing the politician's policy decision. We build on prior political economy models of voter participation in mass events by incorporating a strategic interest group that organizes an event with varying participation cost and also by including in the model a saliency dimension of voter preferences. The model yields novel predictions about how the profile of voters' preferences and how politicians' prior beliefs affect the incentive for an interest group to publicly mobilize its supporters in a low participation-cost event such as an online petition or a higher cost public rally.

Our model provides a foundation for extensions that can address related questions as well as limitations of the current model. A natural extension would be to include an opposing interest group that could mobilize its own supporters in order to counteract the effect of the first interest group's mobilization. Incorporating strategic interactions between two interest groups, voters, and the politician would significantly increase the complexity of the model and the equilibrium conditions. Another extension would be to consider an interest group employing multiple simultaneous types of mobilization campaigns, or even mobilization combined with campaign contributions or other conventional forms of political influence. Exploring strategic actions taken by the interest group that are designed to directly change the profile of stakeholder preferences could also yield new insights. For example, Uber may offer lower fares to its customers or AirBnb may provide better insurance cover to home-sharers in a bid to increase the extent and/or depth of support among its supporters, making the organization of mobilization events more attractive. Such integrated competitive and political strategies have been studied in a few models, though primarily in full information contexts (Baron, 1999, 2001, 2018; De Borger and Glazer, 2015). We leave these and other model enhancements for future research.

Declaration of competing interest

We declare that we have no conflicts of interest related to the co-authored manuscript we are submitting to *Games and Economic Behavior*, "Strategic Mobilization of Voters".

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Appendix A. Supplementary material

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.geb.2025.08.004.

Data availability

No data was used for the research described in the article.

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